# NLP Smart Contract Generator

**Title**
NLP Smart Contract Generator

**Detailed Problem Statement & Requirements:**
NLP Smart Contract Generator focused on bridging natural language and blockchain by converting everyday "If/Then" English descriptions into secure, executable Solidity smart contracts. Participants used LLMs to parse user inputs like "If user deposits 1 ETH, then mint 1000 tokens and lock for 30 days," generating compilable code with automated syntax checks and vulnerability scans. The goal emphasized accessibility for non-devs, targeting DeFi, DAOs, and governance use cases while prioritizing gas efficiency and security.

**Key Requirements**

- Natural language input: Support conditional logic (if/then/else), loops, events, and common patterns (e.g., ERC-20 transfers, access control).
- LLM pipeline: Prompt engineering with OpenAI GPT or LLaMA via LangChain for chain-of-thought reasoning; output raw Solidity + ABI.
- Validation: Integrate Slither/MythX for static analysis (reentrancy, overflow checks); Hardhat for local testing; deployable on Remix.
- Error handling: Fallback to manual edits if generation fails; 95%+ success rate on benchmark prompts.

**Hosting Platform**
ETHGlobal

**Duration**
24hr

**Tech Stack**

- Frontend: Next.js, Tailwind
- Backend: Node.js, Express
- AI: OpenAI / LLaMA, LangChain
- Blockchain: Solidity, Hardhat, Ethers.js, Remix IDE
- Validation: Slither, MythX